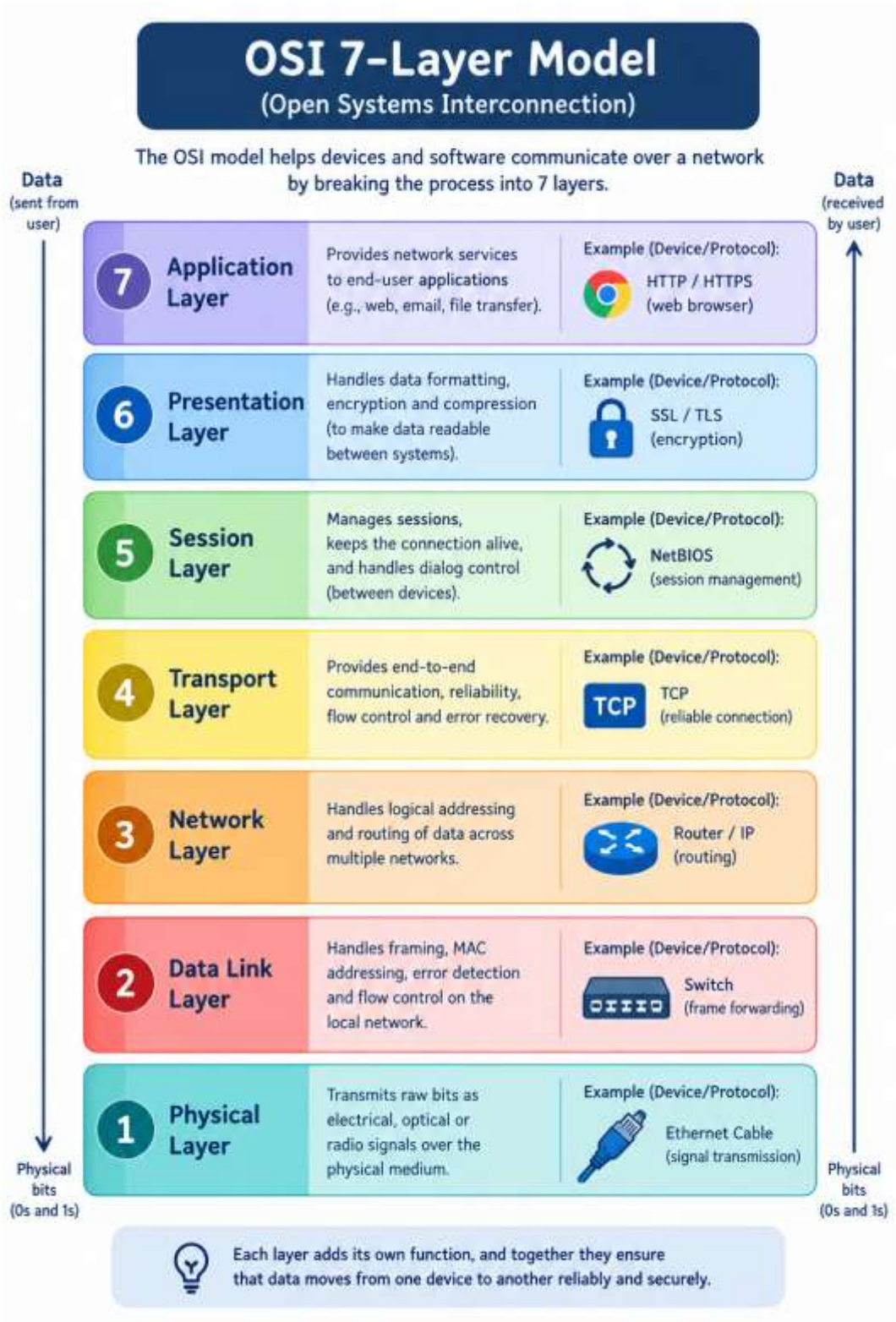
1.

Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).

ANS:-



2.

Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames. Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.

ANS:-

a) Encrypting data before sending: Presentation Layer (Layer 6)

b) Routing packets across networks: Network Layer (Layer 3)

c) Converting data to electrical signals: Physical Layer (Layer 1)

d) Opening a web page in your browser: Application Layer (Layer 7)

e) Detecting and correcting errors in frames: Data Link Layer (Layer 2)

3.

Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

ANS:-

When I send a message on WhatsApp, different OSI layers work together to deliver it.

1. Application Layer (Layer 7): WhatsApp provides the interface to send and receive messages.
2. Transport Layer (Layer 4): Protocols such as TCP or UDP support data transport, depending on the communication service being used.
3. Network Layer (Layer 3): IP addresses and routing help packets travel across networks to reach their destination.
4. Data Link Layer (Layer 2): Transfers data over the local network, such as Wi-Fi, and handles frame-level error detection.
5. Physical Layer (Layer 1): Transmits data as radio signals over Wi-Fi or mobile signals over a cellular connection.

Thus, the OSI layers work together to enable reliable communication between devices.

4.

Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

ANS:-

Possible OSI layers involved:

- Application Layer (Layer 7): The web server or website service may be unavailable or malfunctioning.
- Transport Layer (Layer 4): HTTP/HTTPS connections may fail because TCP ports 80 or 443 are blocked or not responding.
- Application Layer (Layer 7), DNS: DNS resolution may be failing if the domain name cannot be translated into an IP address.

Reasoning: Successfully pinging an IP address indicates that ICMP communication to that IP is working, but it does not guarantee that the website service is working. Check DNS resolution, browser settings, firewall rules, and whether the web server is accepting connections on ports 80 or 443.